

LAB01

Déploiement et exploration d'une infrastructure OpenStack

Guide d'installation pas à pas : DevStack sur une instance AWS EC2

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COURS

Infrastructure Cloud

PLATEFORME

AWS EC2 · DevStack

VERSION

Octobre 2026

01 Provisionnement de l'instance EC2

L'hôte OpenStack est une instance EC2 unique (installation « tout-en-un »). Les paramètres de lancement sont les suivants :

Paramètre	Valeur
Nom	DevStack
AMI	Ubuntu 26.04 LTS
Type d'instance	r8i.large (2 vCPU, 16 Go RAM)
Paire de clés	devstack (RSA, format .pem)
Groupe de sécurité	DevStack SG — SSH (22) et HTTP (80) depuis n'importe où
Stockage	50 Go (gp3)

1.1 Nom et image du système

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
DevStack [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

aws Mac Canonical Ubuntu Microsoft Red Hat SUSE debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Figure 1 — Nom de l'instance (DevStack) et choix de l'AMI Ubuntu

1.2 Type d'instance

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

Q r8i

r8i.large

Family: r8i 2 vCPU 16 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.14242 USD per Hour On-Demand Linux base pricing: 0.13892 USD per Hour

On-Demand SUSE base pricing: 0.19522 USD per Hour On-Demand Windows base pricing: 0.23092 USD per Hour

On-Demand RHEL base pricing: 0.16772 USD per Hour

[All generations](#)

[Compare instance types](#)

Figure 2 — Recherche et sélection du type r8i.large

1.3 Paire de clés

Cliquez sur **Create new key pair**, nommez la clé **devstack**, conservez le type RSA et le format **.pem**, puis créez la clé. Le fichier est téléchargé une seule fois : conservez-le en lieu sûr.

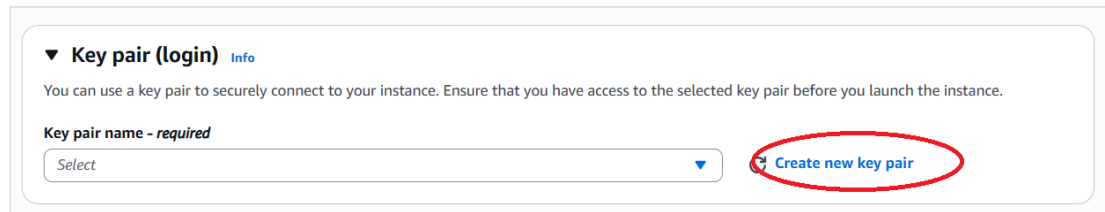


Figure 3 — Lancement de la création de la paire de clés

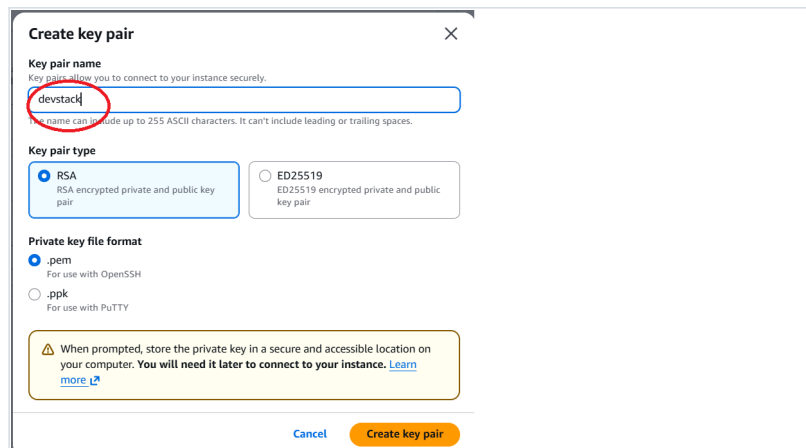


Figure 4 — Paramètres de la clé : nom devstack, RSA, .pem

1.4 Groupe de sécurité

Dans **Network settings**, cliquez sur **Edit**, choisissez **Create security group** et nommez-le **DevStack SG**.

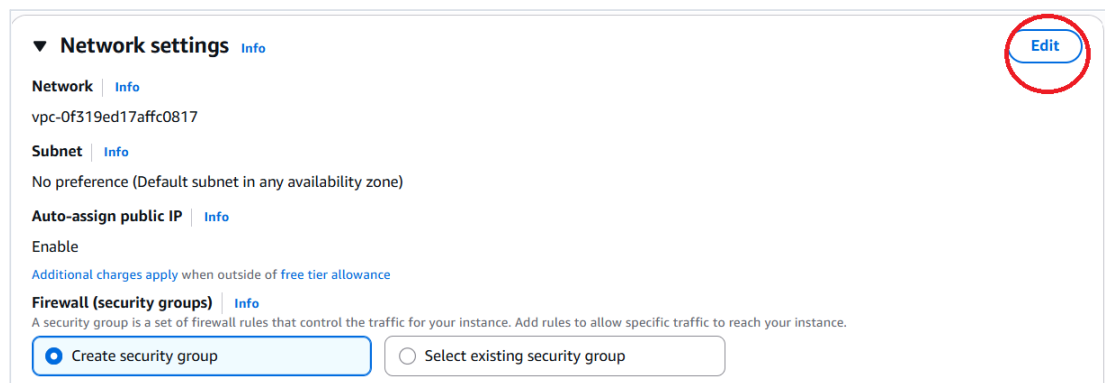


Figure 5 — Accès à l'édition des paramètres réseau

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0f319ed17affc0817 (default) [Refresh](#)

Subnet [Info](#)

No preference [Refresh](#) [Create new subnet](#)

Availability Zone [Info](#)

No preference [Refresh](#) [Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

DevStack SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()#,@!+=&:[]\$*

Description - required [Info](#)

Allow SSH and HTTP access to the DevStack OpenStack host

Figure 6 — Création du groupe de sécurité DevStack SG

Ajoutez ensuite les deux règles entrantes : **SSH** (port 22) puis **HTTP** (port 80), toutes deux avec la source **Anywhere** (`0.0.0.0/0`).

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) [Remove](#)

Type [Info](#) ssh

Protocol [Info](#) TCP

Port range [Info](#) 22

Source type [Info](#) Anywhere

Source [Info](#) [Add CIDR, prefix list or security group](#) 0.0.0.0/0 [X](#)

Description - optional [Info](#) e.g. SSH for admin desktop

► Security group rule 2 (TCP, 0) [Remove](#)

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. [X](#)

[Add security group rule](#)

Figure 7 — Règle 1 : SSH (TCP 22), puis « Add security group rule »

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) Remove

Type: **ssh** | Info | Protocol: **TCP** | Info | Port range: **22** | Info

Source type: **Anywhere** | Info | Source: **0.0.0.0/0** | Info | Description - optional: **e.g. SSH for admin desktop** | Info

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0) Remove

Type: **HTTP** | Info | Protocol: **TCP** | Info | Port range: **80** | Info

Source type: **Anywhere** | Info | Source: **0.0.0.0/0** | Info | Description - optional: **e.g. SSH for admin desktop** | Info

Figure 8 — Règle 2 : HTTP (TCP 80) depuis Anywhere

Bonne pratique de sécurité

Ouvrir SSH à **0.0.0.0/0** est toléré pour un TP à durée de vie courte. Dans un environnement réel, restreignez la source à votre adresse IP.

1.5 Stockage

Réglez la taille du volume racine à **50 Go**, puis cliquez sur **Launch instance**.

▼ **Configure storage** | Info | Advanced

1x **50** GiB | gp3 | Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new | ☐ EFS | ☒ None

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-0fe201115eeef936

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 50 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of...

[Cancel](#) [Launch instance](#)

Figure 9 — Volume racine de 50 Go et lancement de l'instance

02 Affectation de l'adresse IP élastique

Une adresse IP publique standard change à chaque arrêt/démarrage de l'instance. Or l'instance doit être arrêtée à l'étape suivante : on alloue donc une **IP élastique** pour obtenir une adresse stable.

1. Dans le menu de gauche, ouvrez **Network & Security > Elastic IPs**.
2. Cliquez sur **Allocate Elastic IP address**, conservez le pool d'Amazon puis cliquez sur **Allocate**.
3. Sélectionnez l'adresse, puis **Actions > Associate Elastic IP address**.
4. Choisissez l'instance **DevStack** et validez avec **Associate**.

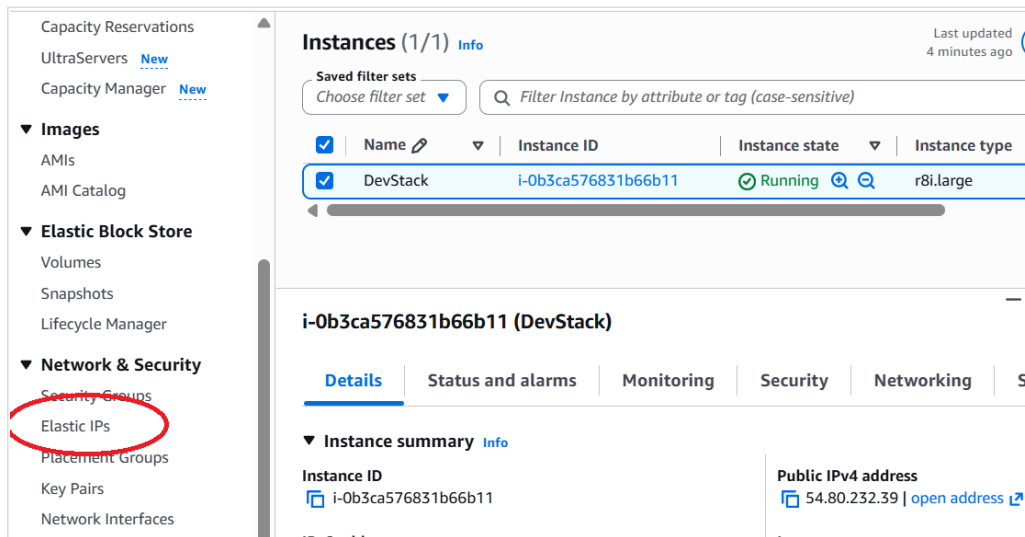


Figure 10 — Menu Elastic IPs de la console EC2

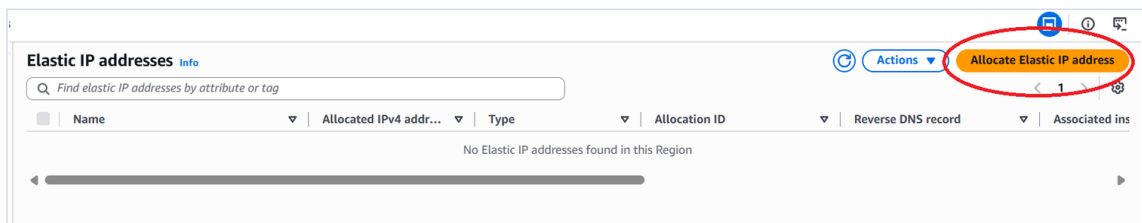


Figure 11 — Allocation d'une nouvelle adresse

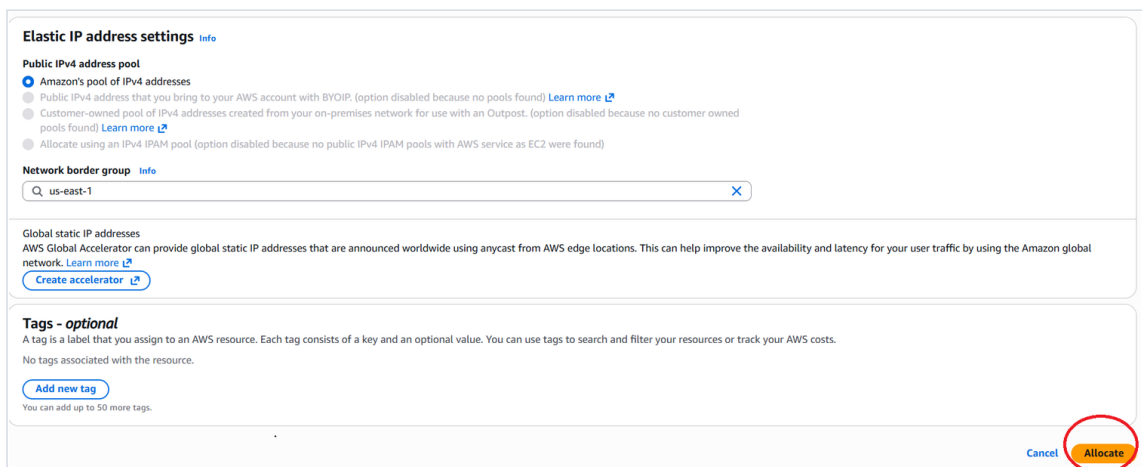


Figure 12 — Paramètres d'allocation (pool Amazon)

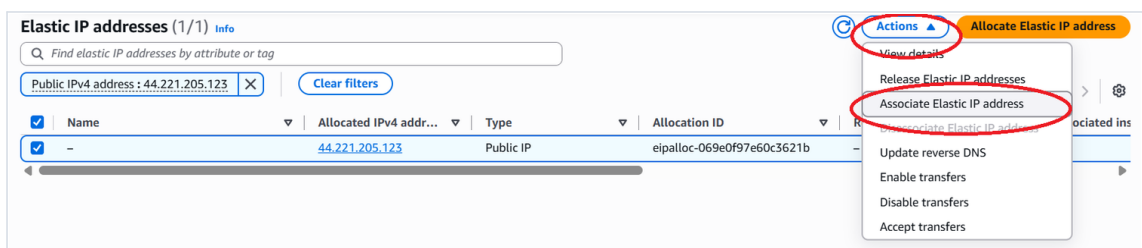


Figure 13 — Actions > Associate Elastic IP address

Associate Elastic IP address [Info](#)
Choose the instance or network interface to associate to this Elastic IP address (44.221.205.123)

Elastic IP address: 44.221.205.123

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

⚠ If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance
Choose an instance

i-0b3ca576831b66b11 (DevStack) - running

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☐ Allow this Elastic IP address to be reassociated

[Cancel](#) [Associate](#)

Figure 14 — Association de l'IP élastique à l'instance DevStack

Attention à la facturation

Une IP élastique et un volume EBS restent facturés même lorsque l'instance est arrêtée. Libérez l'adresse (**Release Elastic IP address**) à la fin du TP.

03 Activation de la virtualisation imbriquée

OpenStack (Nova) délègue la création des VM à KVM, qui requiert les extensions matérielles du CPU (Intel VT-x). L'hôte étant lui-même une VM, AWS doit exposer ces extensions : c'est la **virtualisation imbriquée** (nested virtualization), désactivée par défaut.

1. Sélectionnez l'instance puis **Instance state > Stop instance** (et non Terminate). Attendez l'état **stopped**.
2. Confirmez l'arrêt avec **Stop**.
3. **Actions > Instance settings > Change CPU options**.
4. Dans **Nested virtualization**, choisissez **Enable** puis **Change**.
5. **Instance state > Start instance** pour redémarrer.

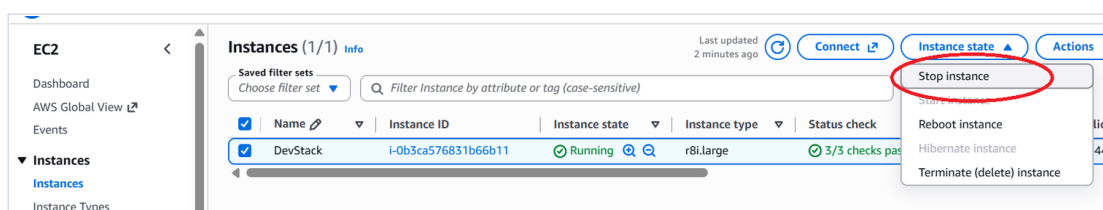


Figure 15 — Instance state > Stop instance

Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID ↗	Stop protection	Result 🚩
i-0b3ca576831b66b11 (DevStack)	Disabled	✔ Can stop

► **Associated resources**
You will continue to incur charges for these resources while the instance is stopped

⚠ **You will be billed for associated resources**
After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

Cancel Stop

Figure 16 — Confirmation de l'arrêt

Instances (1/1) [Info](#)

Connect [↗](#)

Instance state [▼](#)

Actions [▲](#)

Launch instance [➤](#)

Saved filter sets

Choose filter set [▼](#)

Filter Instance by attribute or tag (case-sensitive)

☒	Name ↗	Instance ID	Instance state ▼	Instance type
☒	DevStack	i-0b3ca576831b66b11	Stopped 🔍	r8i.large

i-0b3ca576831b66b11 (DevStack)

Details

Status and alarms

Monitoring

Security

Networking

▼ Instance summary [Info](#)

Instance ID

i-0b3ca576831b66b11

Public IPv4 address

[44.221.205.123](#) [open address](#)

Protection and lifecycle management

Attach to Auto Scaling Group

Change stop protection

Change termination protection

Change shutdown behavior

Change auto-recovery behavior

Change instance migration on reboot

Protect with Elastic Disaster Recovery [↗](#)

Instance configuration

Change instance type

Change CPU options

Modify instance placement

Modify Capacity Reservation settings

Filter actions

Instance diagnostics

Instance settings [▶](#)

Networking [▶](#)

Security [▶](#)

Image and templates [▶](#)

Storage [▶](#)

Monitor and troubleshoot [▶](#)

Figure 17 — Actions > Instance settings > Change CPU options

Change CPU options

Change the number of active vCPUs for your instance to save on licensing costs and optimize performance.

CPU options - update [Info](#)

☐ Use default CPU options

☒ Specify CPU options
Configure CPUs for your instance to optimize performance and save on licensing costs.

Active vCPUs [Info](#)

thread(s) per CPU core across core(s)

Active vCPUs: 2 Total vCPUs: 2

Nested virtualization [Info](#)
Enable nested virtualization to run virtual machines within your instance.

Choose an option [▲](#)

☐ Disable

☒ Enable

Cancel Change

Figure 18 — Nested virtualization : Enable

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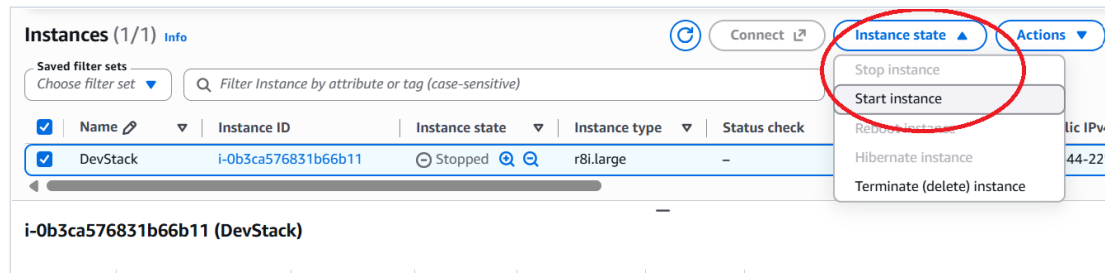


Figure 19 — Redémarrage de l'instance (Start instance)

Vérification

Après redémarrage, `sudo kvm-ok` doit afficher `/dev/kvm exists` et `KVM acceleration can be used`.

04 Connexion SSH

Sélectionnez l'instance, cliquez sur **Connect**, ouvrez l'onglet **SSH client** et copiez la commande proposée.

TERMINAL

```
chmod 400 "devstack.pem"
ssh -i "devstack.pem" ubuntu@<Public_DNS_ou_IP_élastique>
```

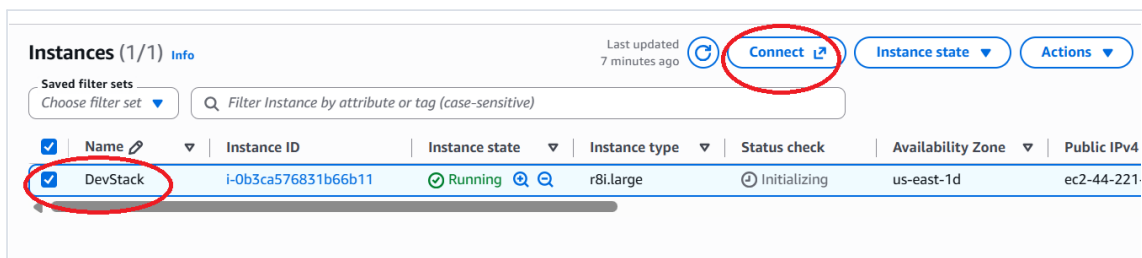


Figure 20 — Bouton Connect de la console EC2

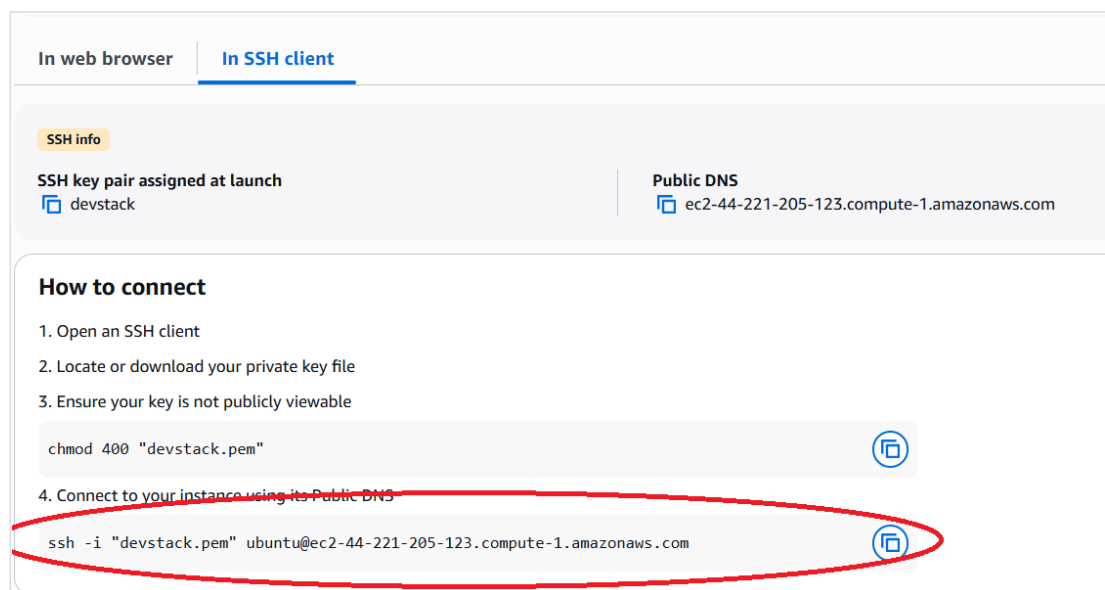


Figure 21 — Onglet « SSH client » et commande de connexion

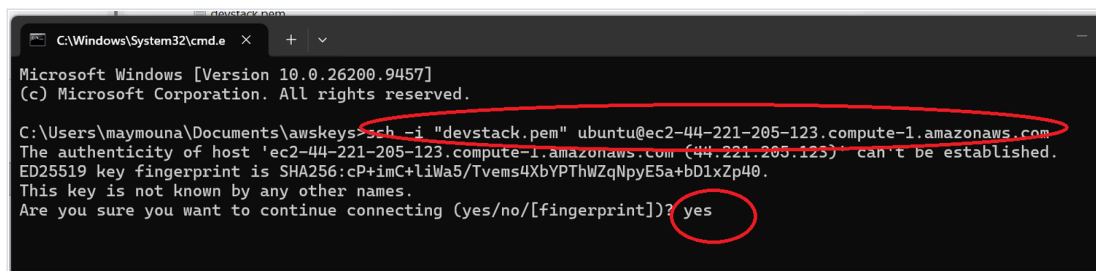


Figure 22 — Première connexion : acceptation de l'empreinte du serveur (yes)

05 Installation de DevStack

Mise à jour du système

TERMINAL

```
sudo apt update
sudo apt upgrade -y
sudo reboot
```

Reconnectez-vous en SSH une fois le redémarrage terminé.

Outils de base et téléchargement de DevStack

TERMINAL

```
sudo apt install -y git curl net-tools
git clone https://opendev.org/openstack/devstack
```

Fichier de configuration local.conf

TERMINAL

```
cd devstack
nano local.conf
```

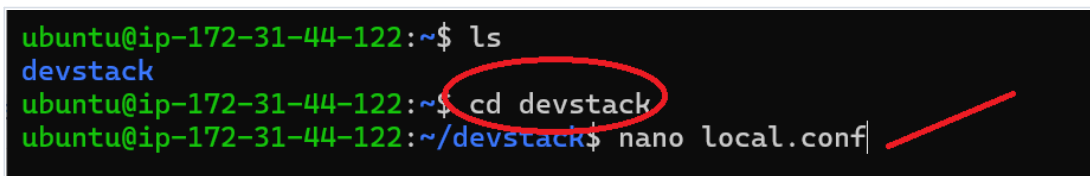


Figure 23 — Création de local.conf dans le dossier devstack

LOCAL.CONF

```
[[local|localrc]]
ADMIN_PASSWORD=secret
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD
```

Mot de passe de démonstration

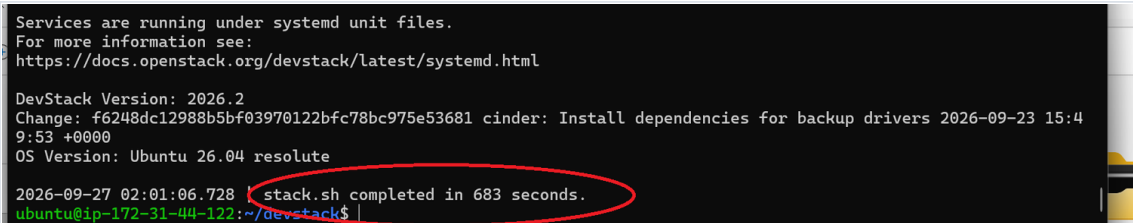
secret convient uniquement pour ce TP. N'utilisez jamais un mot de passe aussi faible sur une plateforme accessible depuis Internet.

Lancement de l'installation

TERMINAL

```
./stack.sh
```

L'installation se termine par le message `stack.sh completed in ... seconds` (683 secondes dans notre cas, soit environ 11 minutes).



```
Services are running under systemd unit files.
For more information see:
https://docs.openstack.org/devstack/latest/systemd.html

DevStack Version: 2026.2
Change: f6248dc12988b5bf03970122bfc78bc975e53681 cinder: Install dependencies for backup drivers 2026-09-23 15:4
9:53 +0000
OS Version: Ubuntu 26.04 resolute

2026-09-27 02:01:06.728 | stack.sh completed in 683 seconds.
ubuntu@ip-172-31-44-122:~/devstack$
```

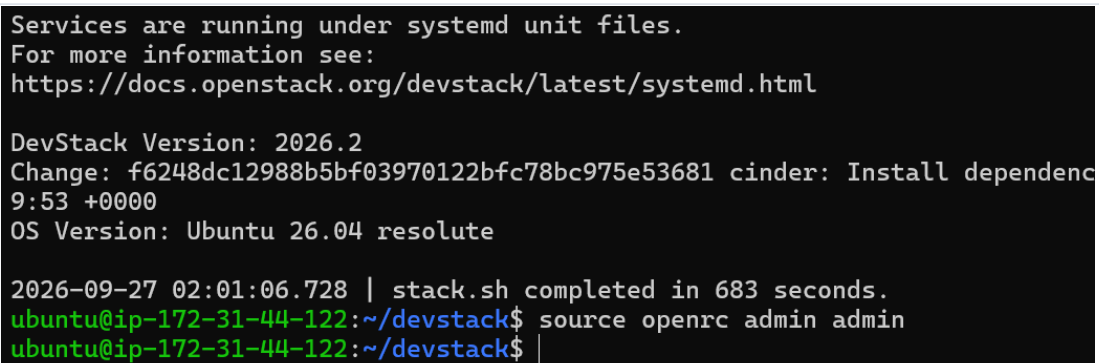
Figure 24 — Fin de l'installation : `stack.sh completed in 683 seconds`

06 Vérification des services (CLI)

Avant d'utiliser la commande `openstack`, chargez dans le shell courant les variables d'environnement d'authentification (Keystone) :

TERMINAL

```
cd ~/devstack
source openrc admin admin
openstack service list
```

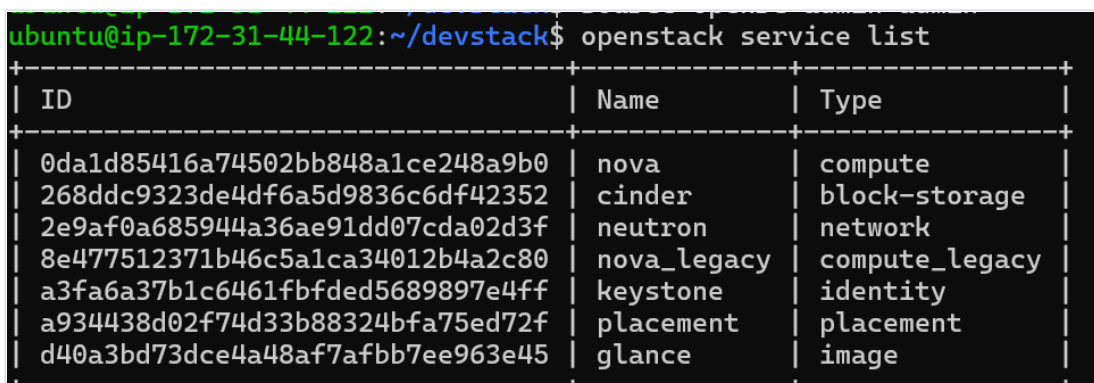


```
Services are running under systemd unit files.
For more information see:
https://docs.openstack.org/devstack/latest/systemd.html

DevStack Version: 2026.2
Change: f6248dc12988b5bf03970122bfc78bc975e53681 cinder: Install dependenc
9:53 +0000
OS Version: Ubuntu 26.04 resolute

2026-09-27 02:01:06.728 | stack.sh completed in 683 seconds.
ubuntu@ip-172-31-44-122:~/devstack$ source openrc admin admin
ubuntu@ip-172-31-44-122:~/devstack$
```

Figure 25 — Chargement des identifiants avec `source openrc admin admin`



```
ubuntu@ip-172-31-44-122:~/devstack$ openstack service list
```

ID	Name	Type
0da1d85416a74502bb848a1ce248a9b0	nova	compute
268ddc9323de4df6a5d9836c6df42352	cinder	block-storage
2e9af0a685944a36ae91dd07cda02d3f	neutron	network
8e477512371b46c5a1ca34012b4a2c80	nova_legacy	compute_legacy
a3fa6a37b1c6461fbfded5689897e4ff	keystone	identity
a934438d02f74d33b88324bfa75ed72f	placement	placement
d40a3bd73dce4a48af7afbb7ee963e45	glance	image

Figure 26 — Résultat de `openstack service list`

Les services suivants doivent apparaître :

Service	Type	Rôle
nova	compute	Création et gestion des machines virtuelles
cinder	block-storage	Volumes de stockage en mode bloc
neutron	network	Réseaux, sous-réseaux, routeurs, IP flottantes
keystone	identity	Authentification et catalogue de services
glance	image	Images de systèmes d'exploitation (ex. cirros)
placement	placement	Inventaire et allocation des ressources
nova_legacy	compute_legacy	Point d'accès de compatibilité de Nova

07 Accès à l'interface graphique Horizon

Élément	Valeur
URL	http://<IP_élastique_de_l_instance>:80
Utilisateur	admin
Mot de passe	secret (défini dans local.conf)

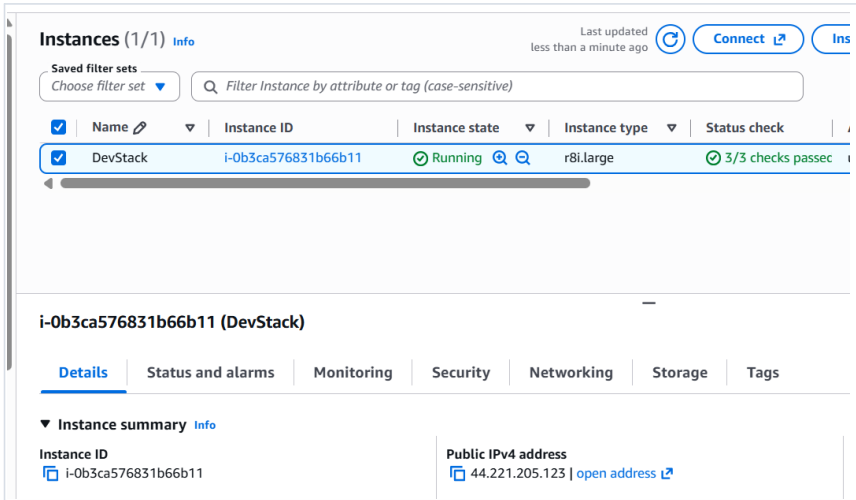


Figure 27 — Instance DevStack en cours d'exécution (3/3 contrôles réussis) et son IP publique

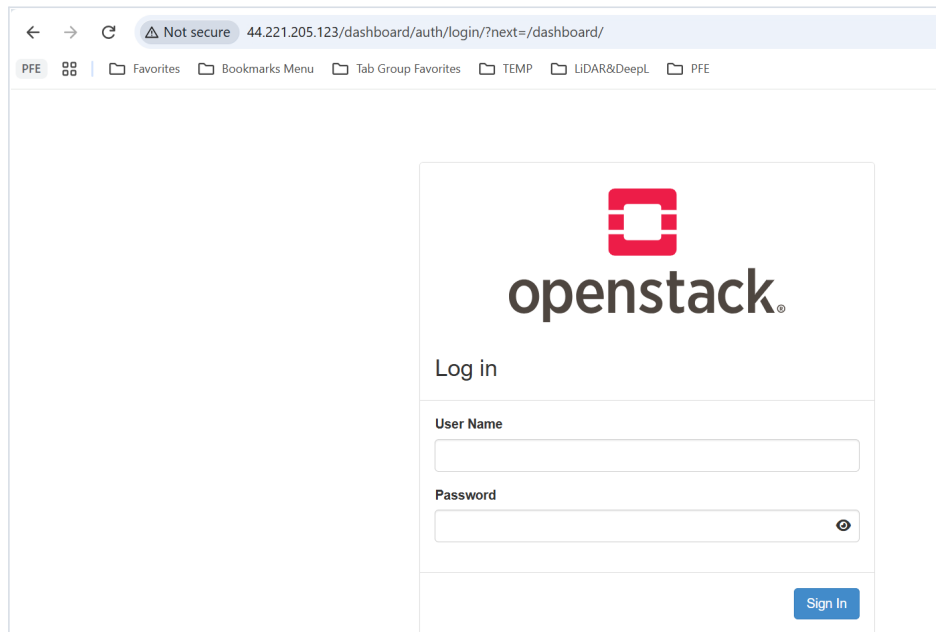


Figure 28 — Page de connexion du dashboard Horizon